

PDF 1 - Flask Fundamentals (10 High-Probability Interview Questions)

1. What is Flask?

Answer: Flask is a lightweight Python web framework used to build web applications and REST APIs. It provides routing, request handling, templating and extensions.

Example / Diagram:

```
from flask import Flask
app = Flask(__name__)
```

Follow-up: Why is Flask called lightweight?

2. Why is Flask called a Micro Framework?

Answer: It provides only the essential web features. Databases, authentication and forms are added through extensions.

Example / Diagram:

```
pip install flask flask-sqlalchemy flask-login
```

Follow-up: What are Flask extensions?

3. Explain `app = Flask(__name__)`

Answer: It creates the Flask application object. `__name__` helps Flask locate templates, static files and resources.

Example / Diagram:

```
app = Flask(__name__)
```

Follow-up: Why not `Flask('app')`?

4. What is WSGI?

Answer: WSGI (Web Server Gateway Interface) is the standard interface between Python web applications and web servers.

Example / Diagram:

```
Browser -> Gunicorn -> Flask App
```

Follow-up: Why is Gunicorn used?

5. What is Werkzeug?

Answer: Werkzeug is the WSGI utility library that powers Flask's routing, request and response handling.

Example / Diagram:

```
from flask import request
```

Follow-up: What features does Werkzeug provide?

6. What is Jinja2?

Answer: Jinja2 is Flask's template engine used to generate dynamic HTML.

Example / Diagram:

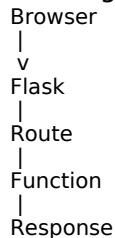
```
return render_template('index.html', name='Sangam')
```

Follow-up: What is template inheritance?

7. Explain the Flask Request-Response Cycle.

Answer: Browser sends request -> Flask matches route -> View function executes -> Response returned.

Example / Diagram:



Follow-up: What happens if no route matches?

8. What does `app.run(debug=True)` do?

Answer: It starts the development server and enables automatic reload and detailed error pages.

Example / Diagram:

```
app.run(debug=True)
```

Follow-up: Should `debug=True` be used in production?

9. Flask vs Django

Answer: Flask is minimal and flexible. Django is full-stack with ORM, admin panel and authentication built in.

Example / Diagram:

```
# Comparison question
```

Follow-up: When would you choose Flask?

10. Development Server vs Production Server

Answer: The built-in Flask server is for development only. In production use Gunicorn/uWSGI behind Nginx.

Example / Diagram:

```
gunicorn app:app
```

Follow-up: Why not use `app.run()` in production?